Quenton Ni

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EDUCATION

University of Minnesota, College of Science and Engineering ***Minneapolis, MN***
M.S. in Computer Science | GPA: 4.00 *May 2027*
B.S. in Computer Science, Minor in Management Information Systems | GPA: 4.00 *May 2026*

EXPERIENCE

IT Intern ***Arden Hills, MN***
Land O' Lakes *May 2026 – Present*
- Architected a web scraping agent using Azure AI Foundry to extract and normalize K-12 meal data across 100+ districts, eliminating an estimated 120 hours of manual sales research while equipping the sales team with new market intelligence.
- Developed vision AI models and automated extraction pipelines for defective pallet claim disputes, aiding manual review to support efforts to recover $3M in annual losses.

Software Engineer (Capstone) ***Minneapolis, MN***
Medtronic *Sep 2025 – May 2026*
- Developed an AI-driven log analysis platform to automatically ingest and process log files from embedded medical devices, streamlining the detection of system anomalies.
- Engineered an automated integration pipeline that correlates detected anomalies with existing bugs, significantly reducing manual overhead.
- Collaborated with Medtronic engineers through the full Agile development lifecycle, ensuring all architectural designs and system components complied with medical technology and regulatory standards.

Research Assistant | *NSF REU & UROP Grant Recipient* ***Minneapolis, MN***
University of Minnesota-Twin Cities *Sep 2025 – May 2026*
- Designed and implemented a Quantization-Aware Training (QAT) pipeline to compress a cardiac signal classification model from FP32 to as low as INT2, achieving up to 30X model size reduction while preserving diagnostic accuracy.
- Engineered Depthwise Separable Convolutional architectures with <400 parameters for intracardiac electrogram (IEGM) classification, targeting deployment on ultra-constrained edge devices.

Computer Science / Machine Learning Teaching Assistant ***Minneapolis, MN***
University of Minnesota-Twin Cities | Data Science and AI Hub *Sep 2024 – May 2026*
- Facilitated the education of 400+ students across Java, C systems programming, and x86-64 assembly by leading weekly labs, holding office hours, and testing course content.
- Supported PhD-led instruction for 90+ high school seniors, teaching core machine learning concepts including deep neural networks, transformers, and reinforcement learning.

PROJECTS

Multi-Dialect Vietnamese Text to Speech
- Fine-tuned and evaluated a multi-dialect Vietnamese TTS system using F5-TTS on ~71 hours of dialect-annotated speech across Northern, Central, and Southern Vietnamese.
- Built and presented an interactive Gradio demo for dialect controlled Vietnamese speech synthesis with generated audio playback and mel spectrogram visualization.

Gamblr
- Engineered a real-time tracking application using Vue.js and Firebase Firestore, featuring a customizable drag anddrop dashboard for live spending analysis and habit tracking.
- Enhanced user interaction by integrating Google Cloud Speech-to-Text via Firebase Functions and rendering an interactive 3D slot machine simulation using Three.js.

Distributed Auction Marketplace System
- Built a container based distributed auction marketplace using Docker Compose, with a centralized controller, stateless service nodes, and replicated SQLite backed storage nodes.
- Implemented primary-backup replication with controller managed health checks and automatic primary failover to maintain availability during storage node failures.
- Designed autoscaling for the service node layer using rolling request rate monitoring and round robin routing.

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, Typescript, HTML, CSS
Technologies: PyTorch, NumPy, Scikit-Learn, Transformers, HuggingFace, Git/GitHub, Docker, Vue, React, Flask, Azure, Azure DevOps, SQL, PostgreSQL, REST APIs, Agile/Scrum
Relevant Coursework: Applied Machine Learning, Natural Language Processing, Algorithms and Data Structures, Operating Systems, Machine Architecture, Distributed Systems